

Dharun Saravanakumar

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EDUCATION

B.Tech in Electronics and Communication Engineering

August 2023 – Present

Indian Institute of Information Technology, Sri City — GPA: 8.4/10.0

Chittoor, AP

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Embedded Systems, IoT, OOP, Computer Networks.

TECHNICAL SKILLS

Programming Languages	C/C++, Python, JavaScript (ES6+), TypeScript, SQL, MATLAB, Verilog
Frameworks & Libraries	React.js, Next.js, Node.js, Express, FastAPI, Flask, Streamlit, TensorFlow, Keras, Scikit-learn, Pandas, OpenCV
Cloud & Database	MongoDB, SQL/NoSQL Databases, Firebase, Google Cloud Platform (GCP), RESTful API Development
Core Competencies	System Design & Architecture, Full-Stack Development, Deep Learning, Data Analysis
Tools & Hardware	Git, Docker, Linux (Bash), VS Code, Raspberry Pi 5, ESP32, Arduino, Unity3D
Soft Skills	Problem Solving, Team Collaboration, Technical Documentation, Analytical Thinking

PROFESSIONAL EXPERIENCE

Titan Company Limited - Software Engineer Intern

June 2025 – July 2025

- Architected a scalable full-stack system for enterprise HR operations using the **MERN stack**, enhancing internal data processing efficiency by **25%**.
- Secured sensitive infrastructure by implementing **JWT-based authentication** and robust RESTful APIs for **500+ active directory users**.
- Optimized frontend performance by **30%** through React-based dashboards featuring advanced state management and data-driven server-side filtering.
- Engineered integrated **attendance and project management modules** with secure file handling, reducing manual administrative overhead by **40%**.

Titan Company Limited - Frontend Developer

June 2024 – June 2024

- Developed a responsive vendor partnership landing page using **HTML, CSS, and ES6+ JavaScript**, collaborating with the **Python/Django backend team** for integration.
- Implemented interactive UI components and ensured **cross-browser compatibility**, supporting the **MBR Digital Transformation** initiative.

TECHNICAL PROJECTS

Trinetra Rover – Multi-Modal AI Surveillance Platform

August 2025 – December 2025

- Spearheaded a semi-autonomous rover that secured **external research funding**, using **Raspberry Pi 5** for edge AI and **ESP32** for sensor fusion.
- Deployed an **Acoustic Classification** model and a **YOLO Visual AI** pipeline, achieving **92% classification accuracy** in identifying threats.
- Developed a high-performance real-time dashboard using **React.js** and **Firebase**, achieving **sub-200ms latency** via UDP command protocols.

Formix – Conversational Form Intelligence & SaaS Platform

Aug 2026 – Present

- Architected a full-stack SaaS platform using **Next.js 16 (App Router)**, **React 19**, **TypeScript**, **FastAPI**, and **SQLAlchemy 2.0 (PostgreSQL/SQLite)**, featuring drag-and-drop form building and keyboard-driven animated respondent flows.
- Designed an atomic debounced autosave engine with temporary ID reconciliation via **TanStack Query v5**, and integrated **Ask Formix AI (Gemini 2.0 Flash / Groq / OpenAI)** to auto-generate form schemas and analyze respondent sentiment.
- Engineered an event-driven **Automations Engine** supporting live webhook execution, conditional action routing logic, aggregate CSV analytics, and an auto-syncing CRM **Contacts Hub**.

FreshAudit – B2B Produce Supply Chain & Cold-Chain Platform

Jan 2026 – Mar 2026

- Built a full-stack B2B SaaS platform using **React 19**, **Vite**, **Node.js**, **Express**, and **Firebase Realtime DB**, with JWT authentication, role-based access control, and chain-of-custody tracking.
- Implemented an **Arrhenius kinetic decay physics engine** over live IoT sensor telemetry for shelf-life prediction, and integrated **Leaflet.js** for live GPS transit mapping with a dynamic pricing engine.

LEADERSHIP & ACHIEVEMENTS

- Web Development Lead, GDG (Google Developer Group) IIIT Sri City** (2025 – Present): Directing technical mentorship for **750+ students** in modern web stacks and AI-driven application deployment through workshops and bootcamps.
- Electroforge Hackathon Winner** (November 2024): Secured **1st place** among 50+ teams for a real-time IoT water-quality monitoring system with a React-based analytics dashboard.
- Core Sponsorship Team – Abhisarga '25** (Annual Cultural Fest): Negotiated corporate partnerships, contributing to a **20% increase** in the flagship event's operational budget.